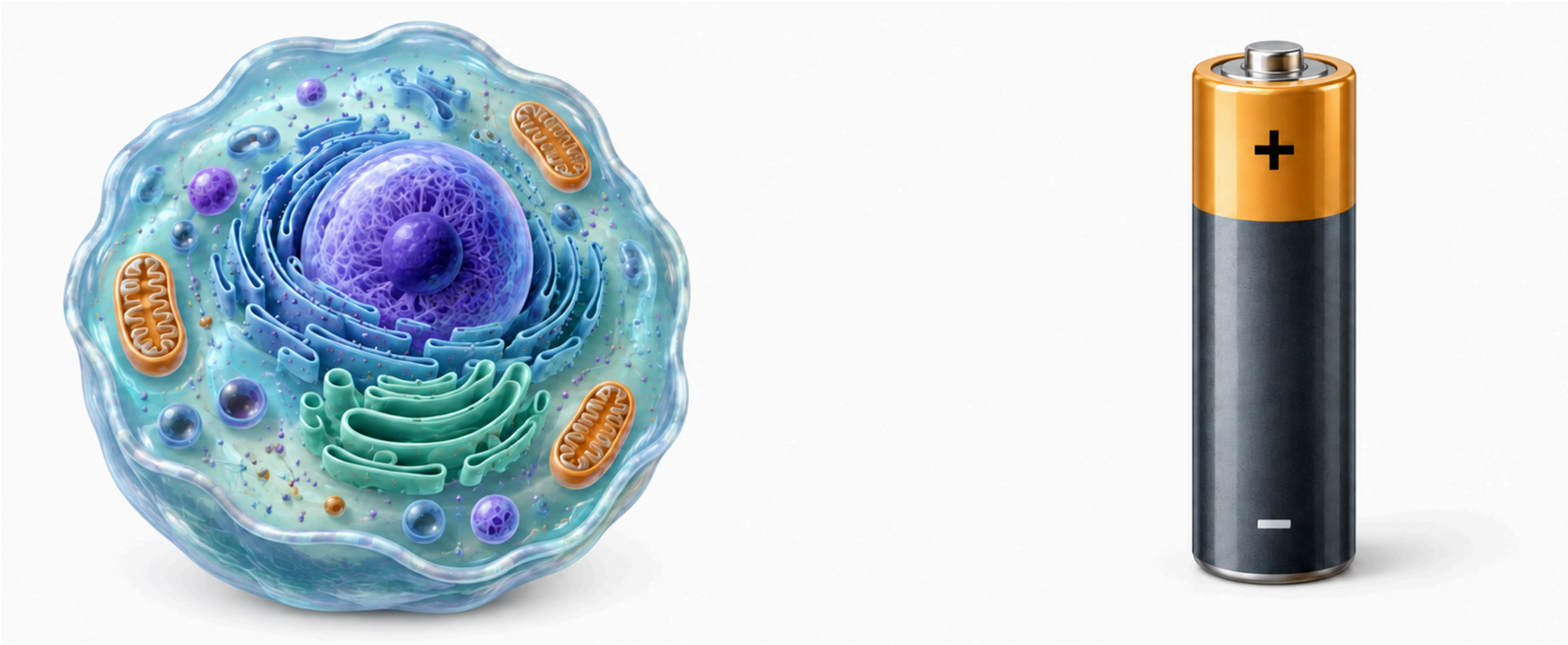
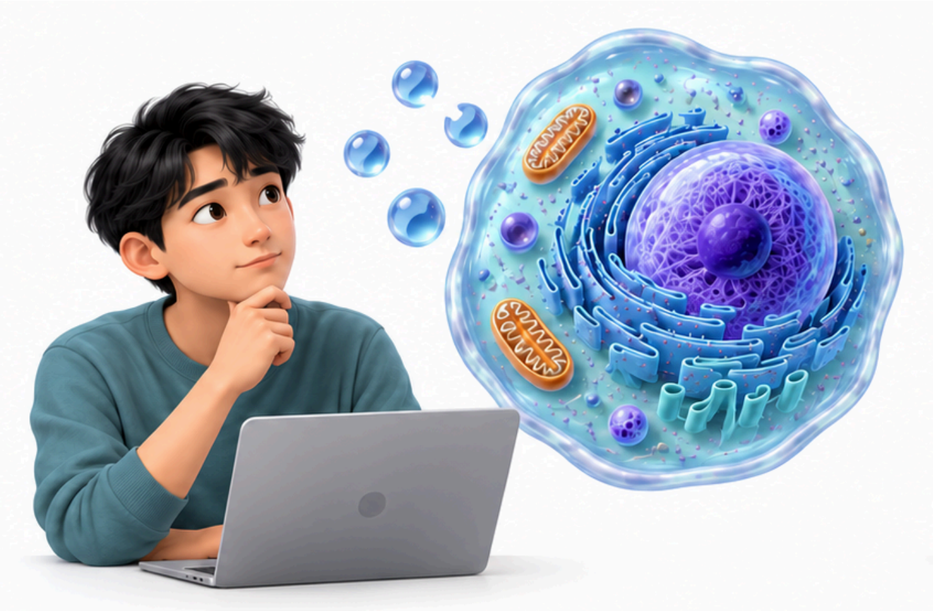
# What does “Cell” mean here?



Biology

Electricity

**How does water enter a cell?**



Concept Explanation

**Can you explain it differently?**



Further Help

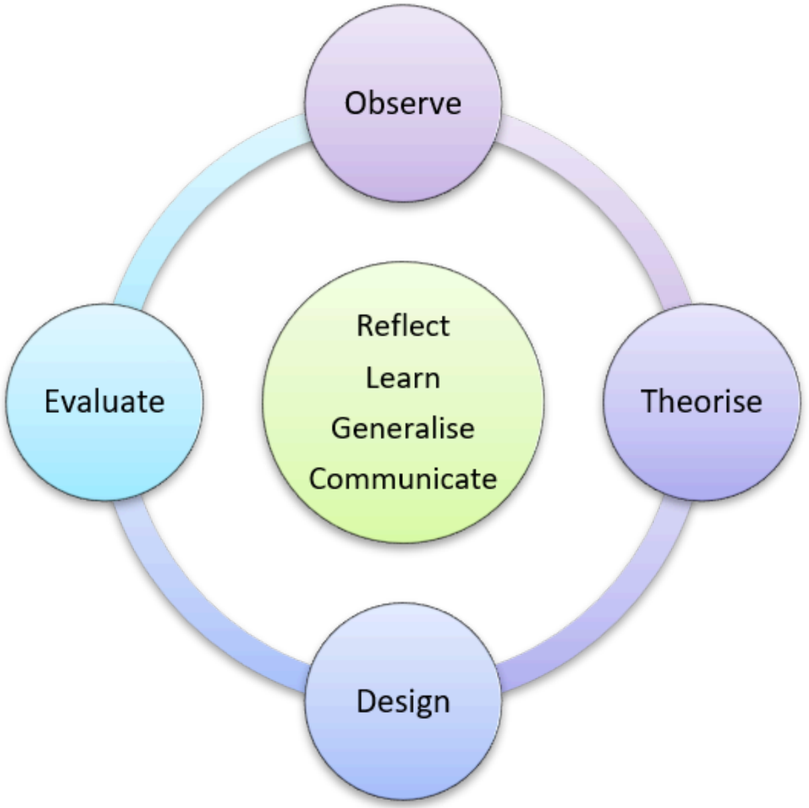
# Adaptive LLM Scaffolding for Burmese-Speaking STEM Learners

Terminology  Concept Explanation  Adaptive help

**Nay Lin Htet (Nathan)**

**University of Auckland, INFOSYS 720**

# RESEARCH METHOD




Information System Research Process
(Gregor & Hevner, 2013)

- **Approach**: Design Science Research (DSR) (Hevner et al., 2004)
- **Main method**: Systems development (Nunamaker et al., 1990)
- **Purpose**: Develop a solution and reusable design guidance

**OBSERVATION**

THEORISING

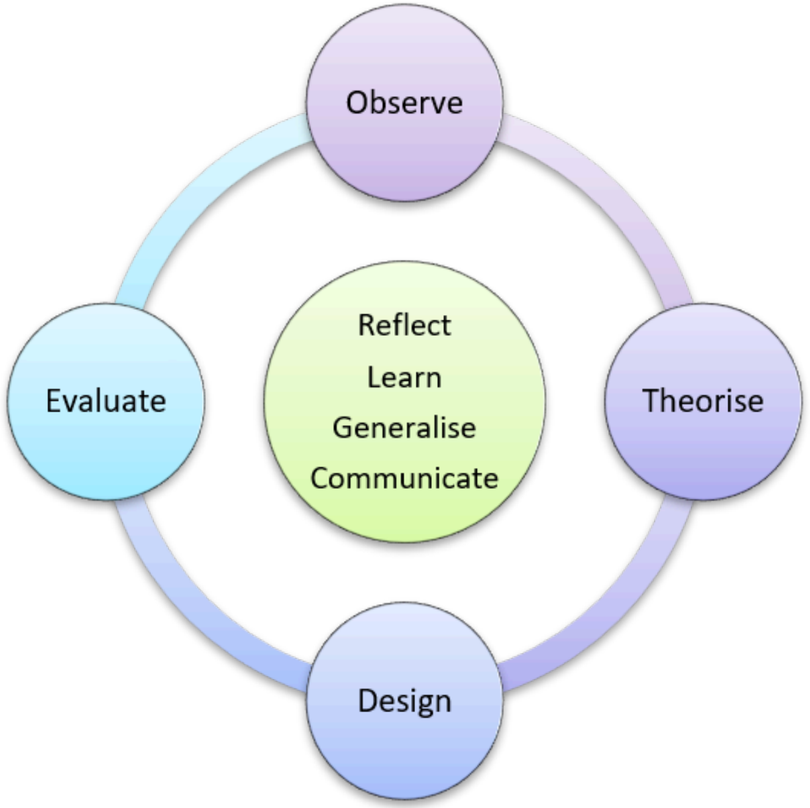
DESIGNING

EVALUATION

REFLECTION

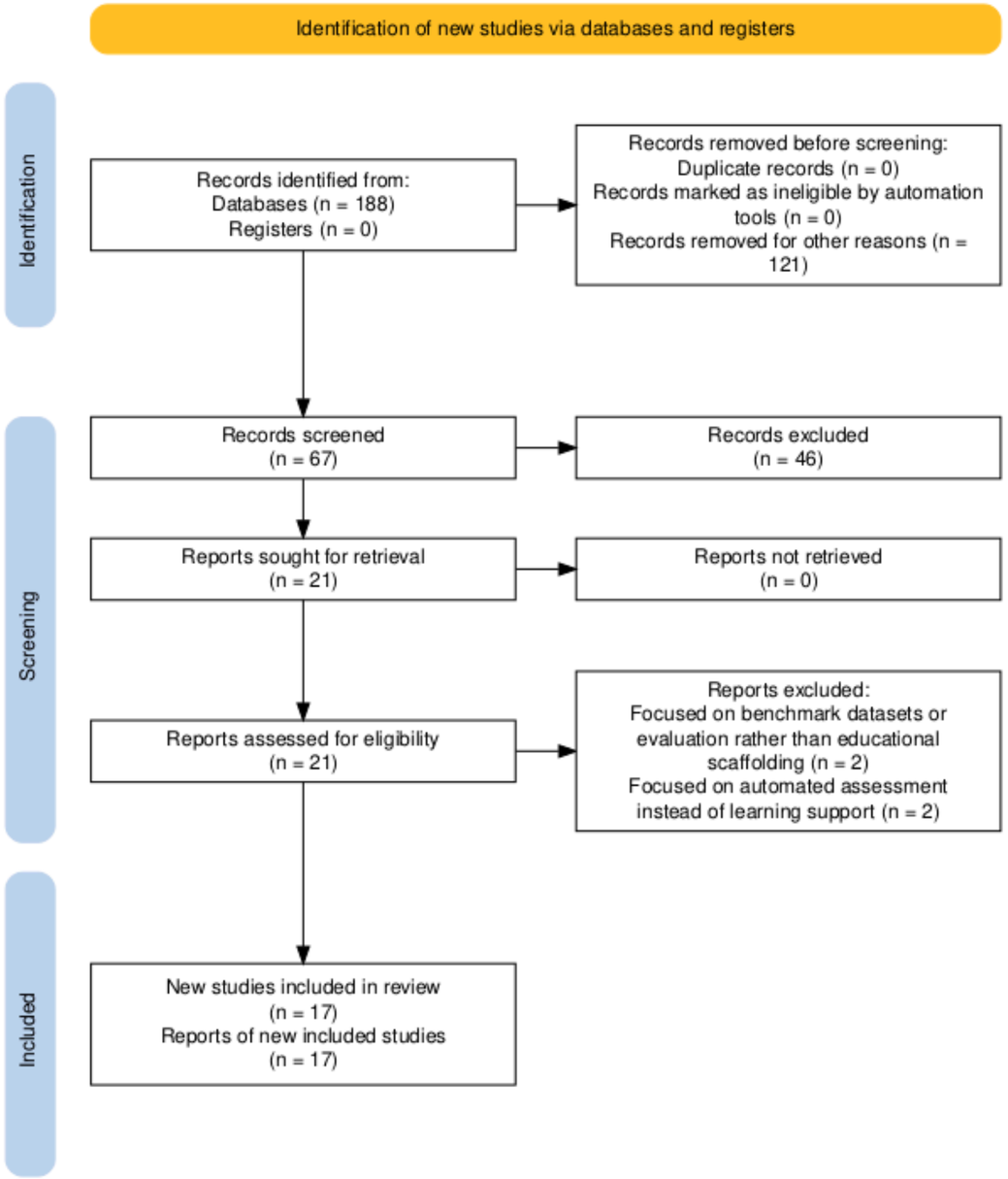
GENERALISATION

CONTRIBUTION




Information System Research Process
(Gregor & Hevner, 2013)

# OBSERVATION




**PRISMA: Adapted from Page et al. (2021)**

- Found 188 records in Scopus (8 Aug 2026)
- Removed 121 records by open-access filter
- Screened 67 records.
  - Excluded 46 for insufficient relevance.
- Full-text reviewed 21 records. 4 excluded.
  - 2 focused on benchmark and evaluation.
  - 2 focused on automated assessment.
- Included 17 studies

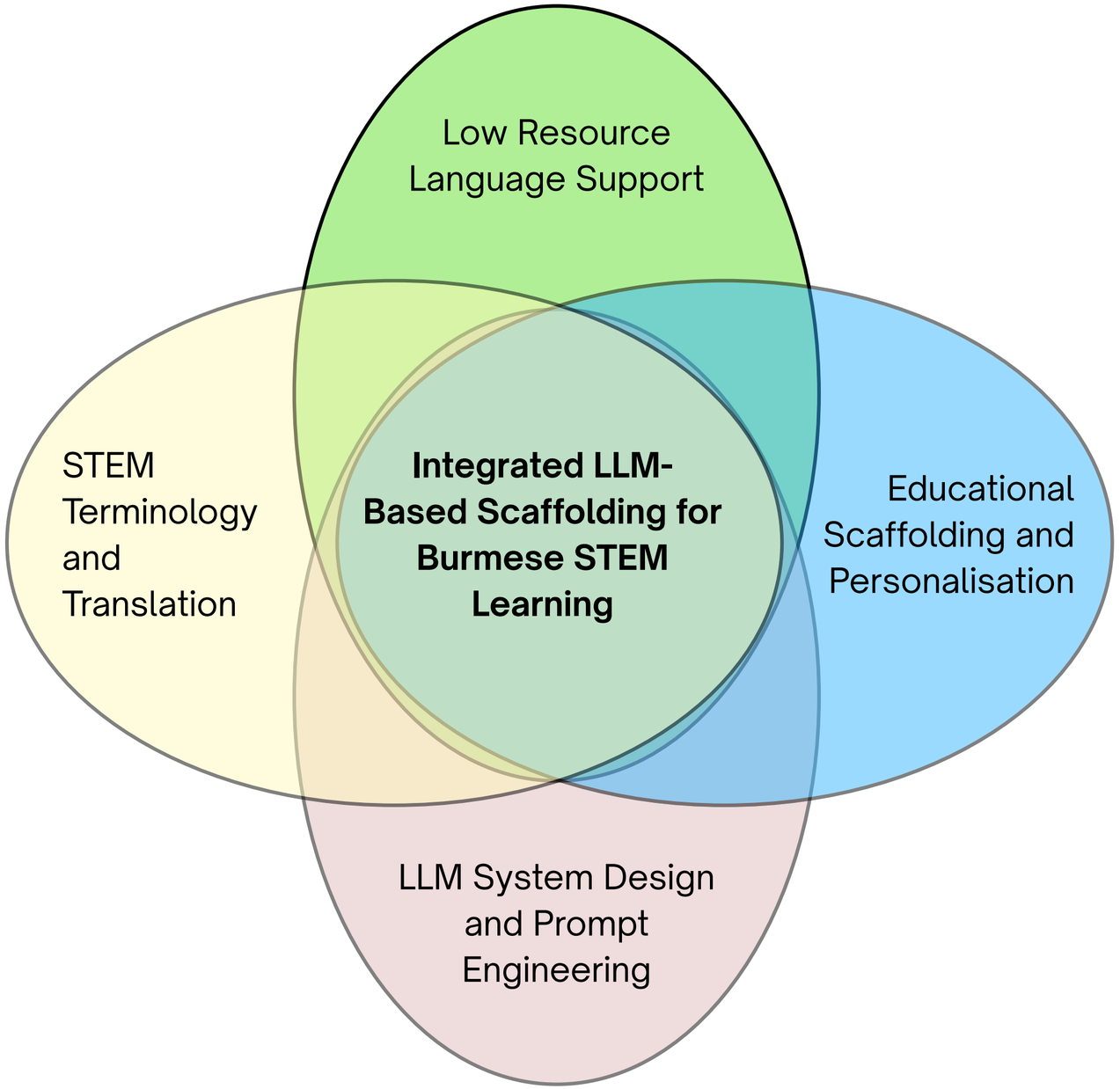
# OBSERVATION

- Topic coverage varied across the reviewed studies
- No reviewed study strongly covered all 12 sub-themes

| Authors | Low Resource and Multilingual Language Barriers | | | LLM-Based Translation and Terminology Support | | | Educational Scaffolding and Personalized Learning | | | | System Design and Prompt Engineering | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Low Resource Language Challenges | Multilingual Learning Barriers | Cultural and Linguistic Context | Context-Aware Translation | Technical Terminology | Terminology Identification | Conceptual Explanation | Personalised / Adaptive Support | Multilingual Interaction | Structured Learning Support | Prompt Engineering | Integrated Educational System |
| He et al. (2025) | 2 | 2 | 3 | 3 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 |
| Bal and Mandal (2026) | 3 | 1 | 3 | 2 | 0 | 0 | 0 | 0 | 1 | 0 | 3 | 0 |
| Kuzu (2026) | 1 | 3 | 2 | 1 | 0 | 0 | 3 | 1 | 3 | 3 | 3 | 2 |
| Zhang and Pang (2025) | 0 | 1 | 0 | 1 | 0 | 0 | 2 | 1 | 1 | 3 | 1 | 3 |
| Candé and Martinho (2026) | 3 | 3 | 3 | 3 | 2 | 0 | 2 | 2 | 2 | 2 | 1 | 1 |
| Athukorala and De Silva (2025) | 3 | 3 | 1 | 3 | 3 | 1 | 3 | 1 | 2 | 3 | 1 | 3 |
| Yu et al. (2025) | 1 | 3 | 2 | 2 | 1 | 0 | 2 | 3 | 3 | 3 | 1 | 1 |
| Rakhimova et al. (2024) | 3 | 2 | 2 | 2 | 1 | 1 | 2 | 1 | 1 | 3 | 0 | 3 |
| Ataman et al. (2025) | 3 | 1 | 2 | 3 | 2 | 1 | 0 | 0 | 0 | 0 | 2 | 0 |
| Vatsal et al. (2026) | 2 | 1 | 2 | 2 | 1 | 0 | 0 | 0 | 0 | 0 | 3 | 0 |
| Khoboko et al. (2025) | 3 | 0 | 1 | 3 | 2 | 1 | 0 | 0 | 0 | 0 | 3 | 0 |
| Tran et al. (2026) | 1 | 0 | 0 | 0 | 3 | 3 | 0 | 0 | 0 | 0 | 1 | 0 |
| Kleidermacher and Zou (2026) | 2 | 1 | 1 | 3 | 3 | 1 | 1 | 0 | 1 | 0 | 3 | 0 |
| Sakunkoo et al. (2025) | 0 | 2 | 2 | 3 | 1 | 0 | 1 | 2 | 3 | 2 | 2 | 3 |
| Bo et al. (2025) | 1 | 2 | 2 | 2 | 1 | 0 | 2 | 3 | 2 | 2 | 1 | 0 |
| Yüksel et al. (2024) | 2 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 |
| Nair et al. (2026) | 3 | 3 | 3 | 2 | 0 | 0 | 2 | 3 | 3 | 3 | 3 | 3 |

0 = No coverage, 1 = Limited, 2 = Moderate, 3 = Strong; darker red indicates stronger coverage

# OBSERVATION




**Venn Diagram**

OBSERVATION

**THEORISING**

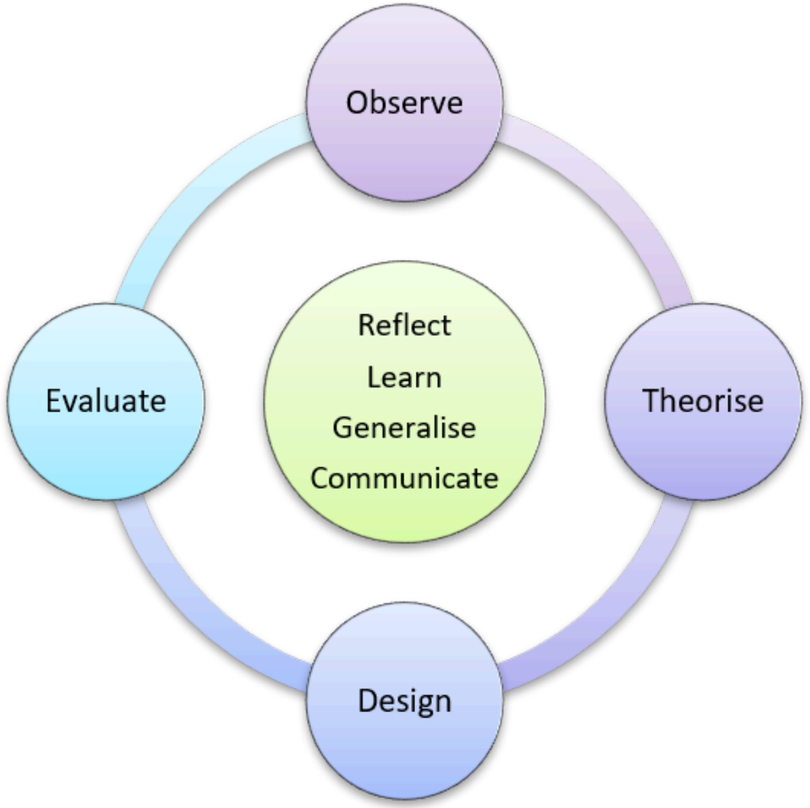
DESIGNING

EVALUATION

REFLECTION

GENERALISATION

CONTRIBUTION




Information System Research Process
(Gregor & Hevner, 2013)

# THEORISING



## Task–Technology Fit

Goodhue and Thompson (1995)

A Burmese-speaking student wants to understand “photosynthesis”.

- Extract the English term and Explain in Burmese.

Photosynthesis ဆိုတာ အပင်တွေက နေရောင်ခြည်၊ ရေ၊ နဲ့ carbon dioxide ကို သုံးပြီး ကိုယ်တိုင် အစာလုပ်တဲ့ လုပ်ငန်းစဉ်ပါ။



## Scaffolding Theory

van de Pol et al. (2010)

Student says “I still need help”.

- Give a simpler rice-plant example.

စပါးရွက်ကို အစာချက်တဲ့ မီးဖိုချောင်လေးလို စဉ်းစားကြည့်ပါ။ အမြစ်က စုပ်ယူလာတဲ့ ရေနဲ့ လေထဲက carbon dioxide ကို နေရောင်ခြည်စွမ်းအင်သုံးပြီး စပါးပင်ကြီးထွားဖို့ လိုတဲ့ အစာဖြစ်တဲ့ သကြားအဖြစ် ပြောင်းလဲပေးပါတယ်။

# THEORISING



## RQ1: Language Barrier

How can specialized English STEM terminology be supported for Burmese-speaking learners?

**Objective**: Develop an approach for **multilingual STEM terminology support**.



## RQ2: Conceptual Difficulties

How can LLM-based support help learners understand STEM concepts beyond translation?

**Objective**: Develop an approach for **conceptual STEM explanation**.



## RQ3: Adaptive Guidance

How can LLM-based scaffolding provide structured and adaptive support for Burmese-speaking STEM learners?

**Objective**: Develop an **adaptive scaffolding approach**.

OBSERVATION

THEORISING

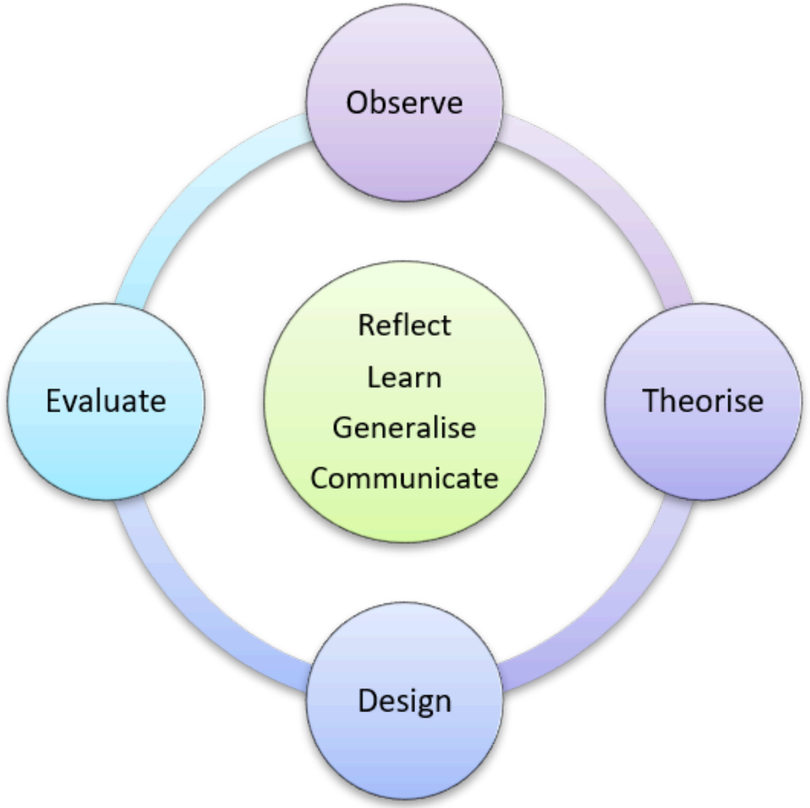
**DESIGNING**

EVALUATION

REFLECTION

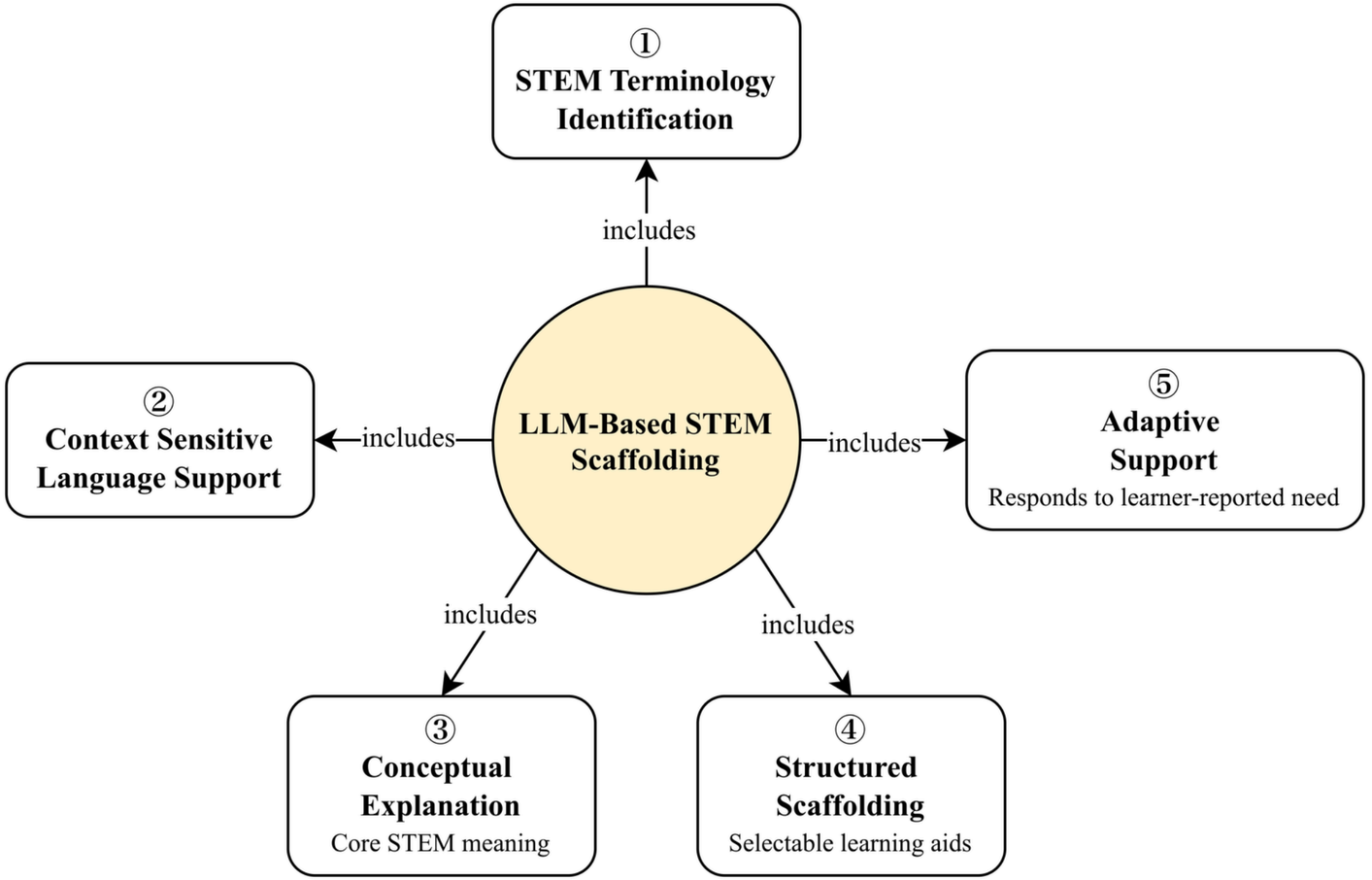
GENERALISATION

CONTRIBUTION




Information System Research Process
(Gregor & Hevner, 2013)

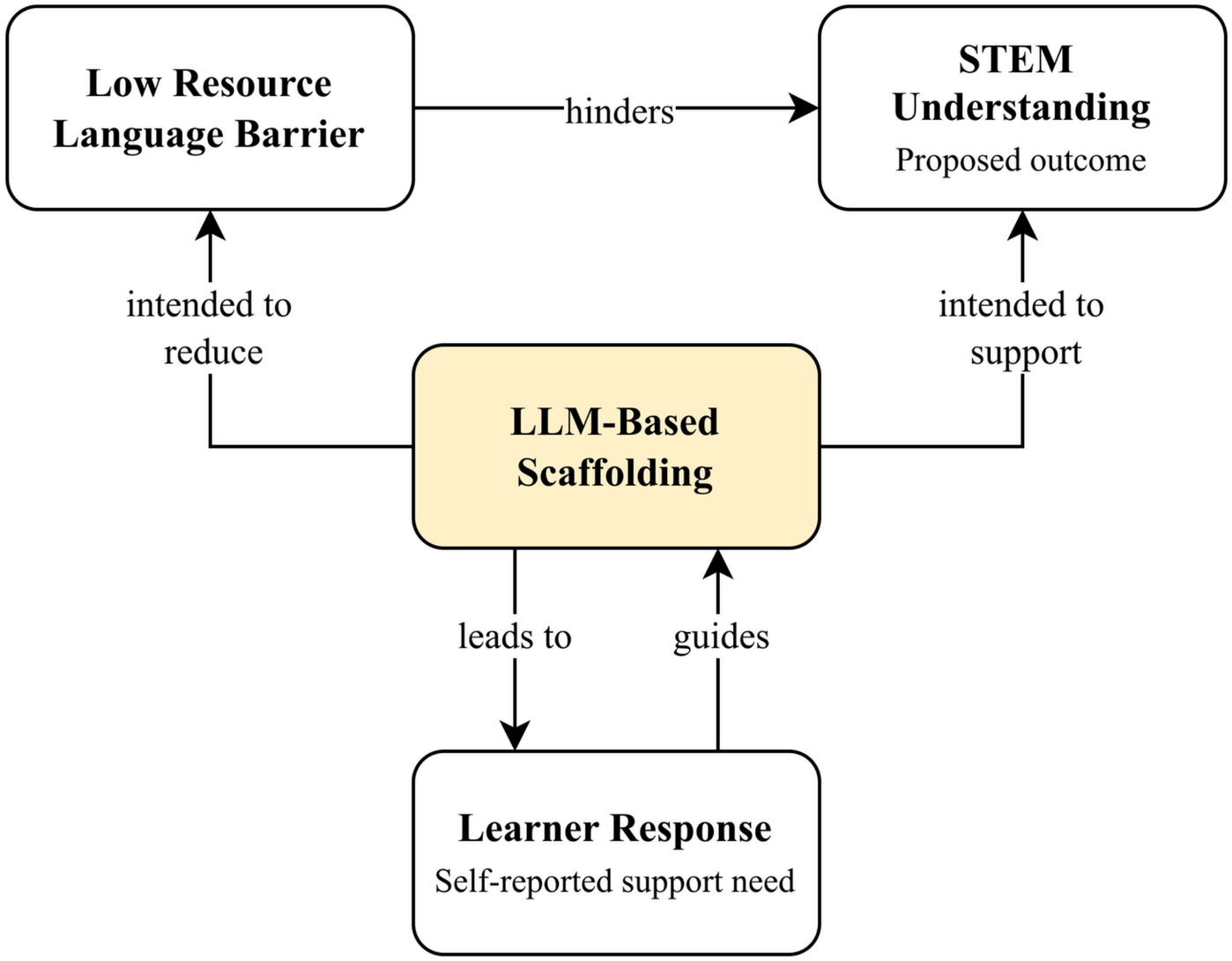
# DESIGNING: CONCEPTUAL ARTEFACTS




**Conceptual Artefact 1**

Conceptual Structure of LLM-Based STEM Scaffolding

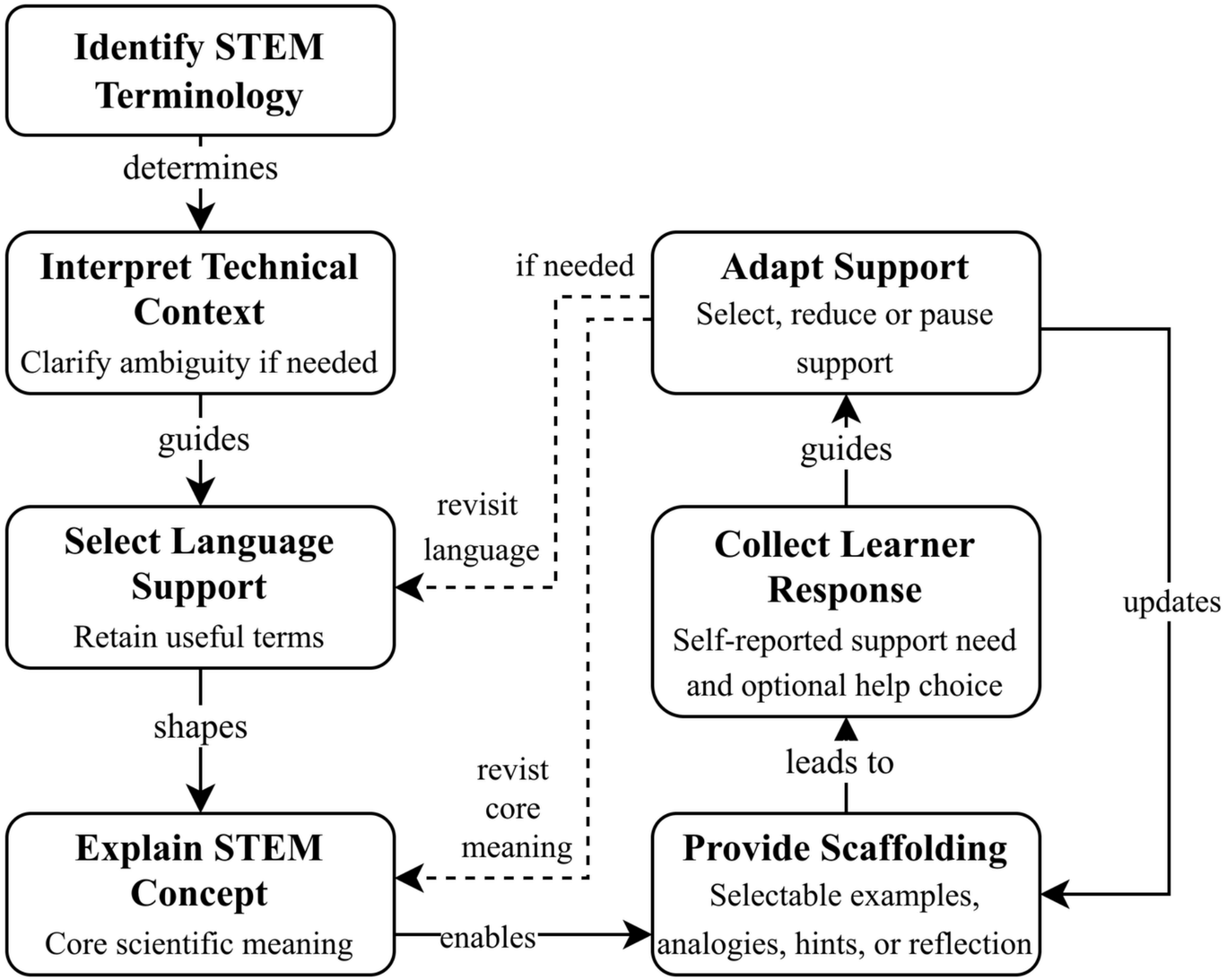
# DESIGNING: CONCEPTUAL ARTEFACTS




**Conceptual Artefact 2**

Task-Aligned LLM Scaffolding Model

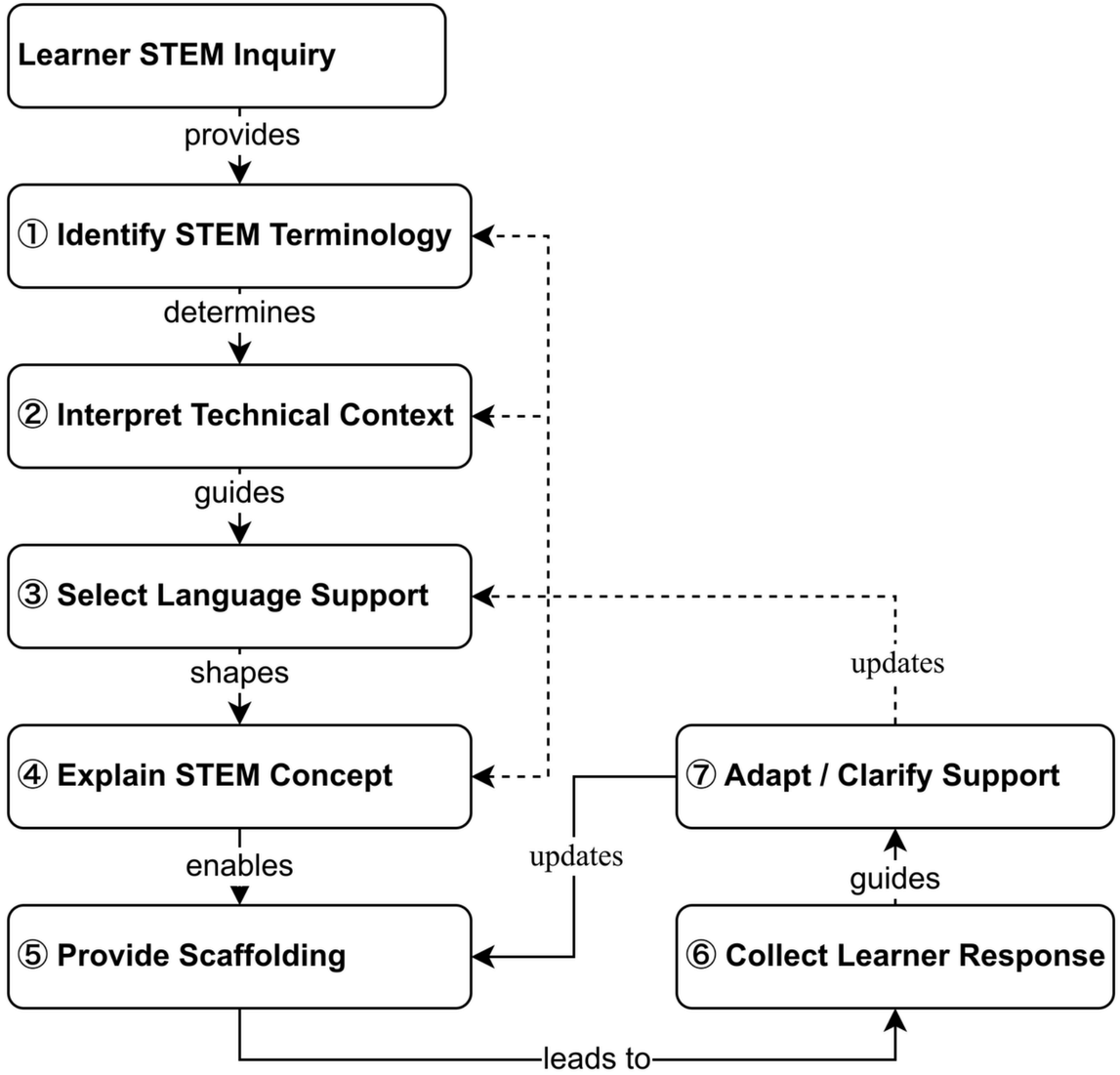
# DESIGNING: CONCEPTUAL ARTEFACTS




**Conceptual Artefact 3**

Context-Aware Adaptive STEM Scaffolding Framework

# DESIGNING: SYSTEM ARTEFACTS

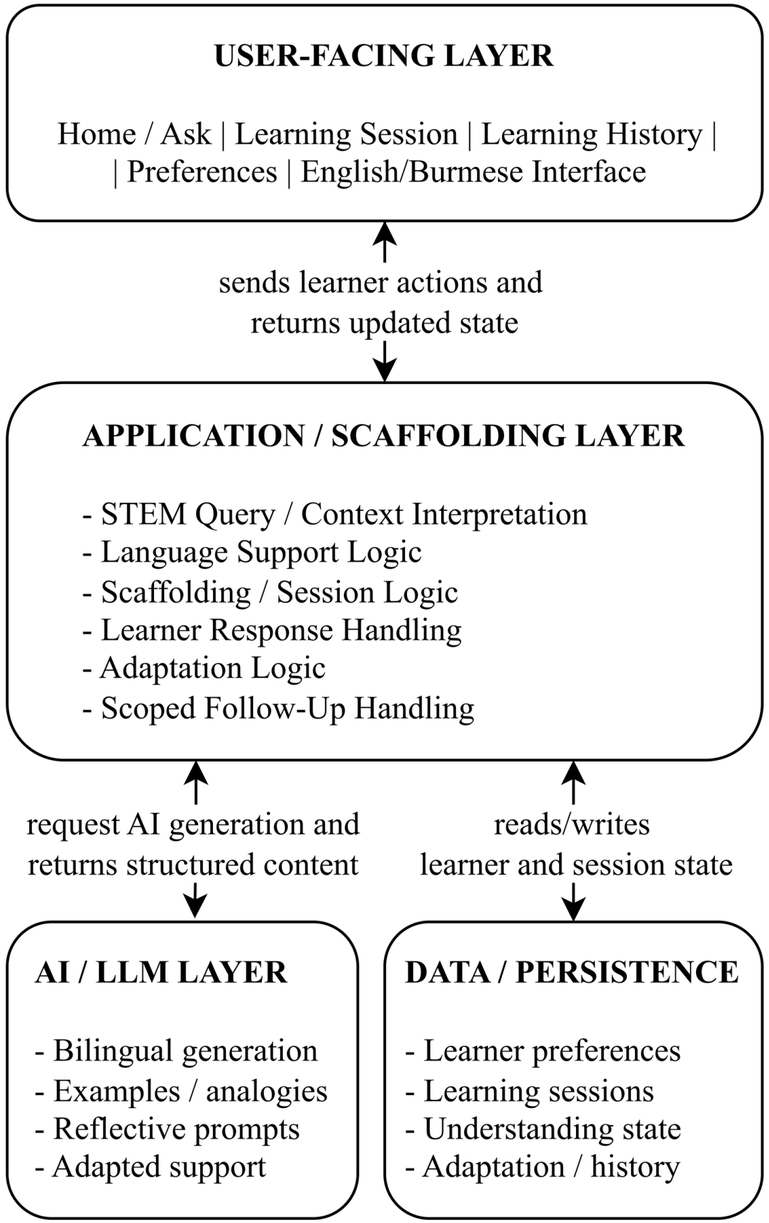



**System Artefact 1**

Adaptive STEM Scaffolding System Framework

- Step 1, 2, 3 address **RQ1: Terms, context and language**.
- Step 4, 5 address **RQ2: Explanation and scaffolding.**
- Step 6, 7 address **RQ3: Learner response and adaptation**.

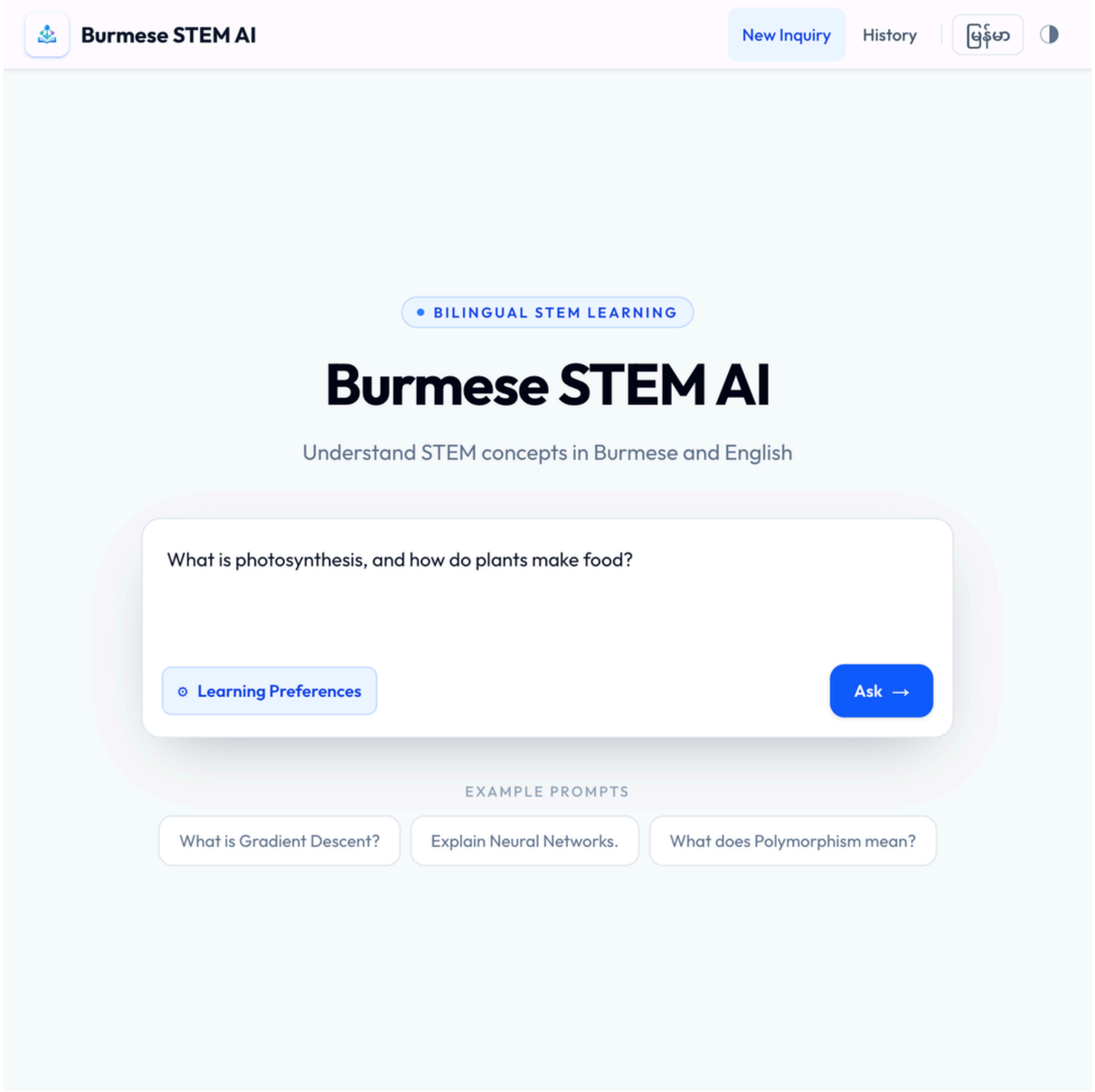
# DESIGNING: SYSTEM ARTEFACTS




**System Artefact 2**

Adaptive STEM Scaffolding Logical Architecture and Information Flow

# DESIGNING: SYSTEM ARTEFACTS



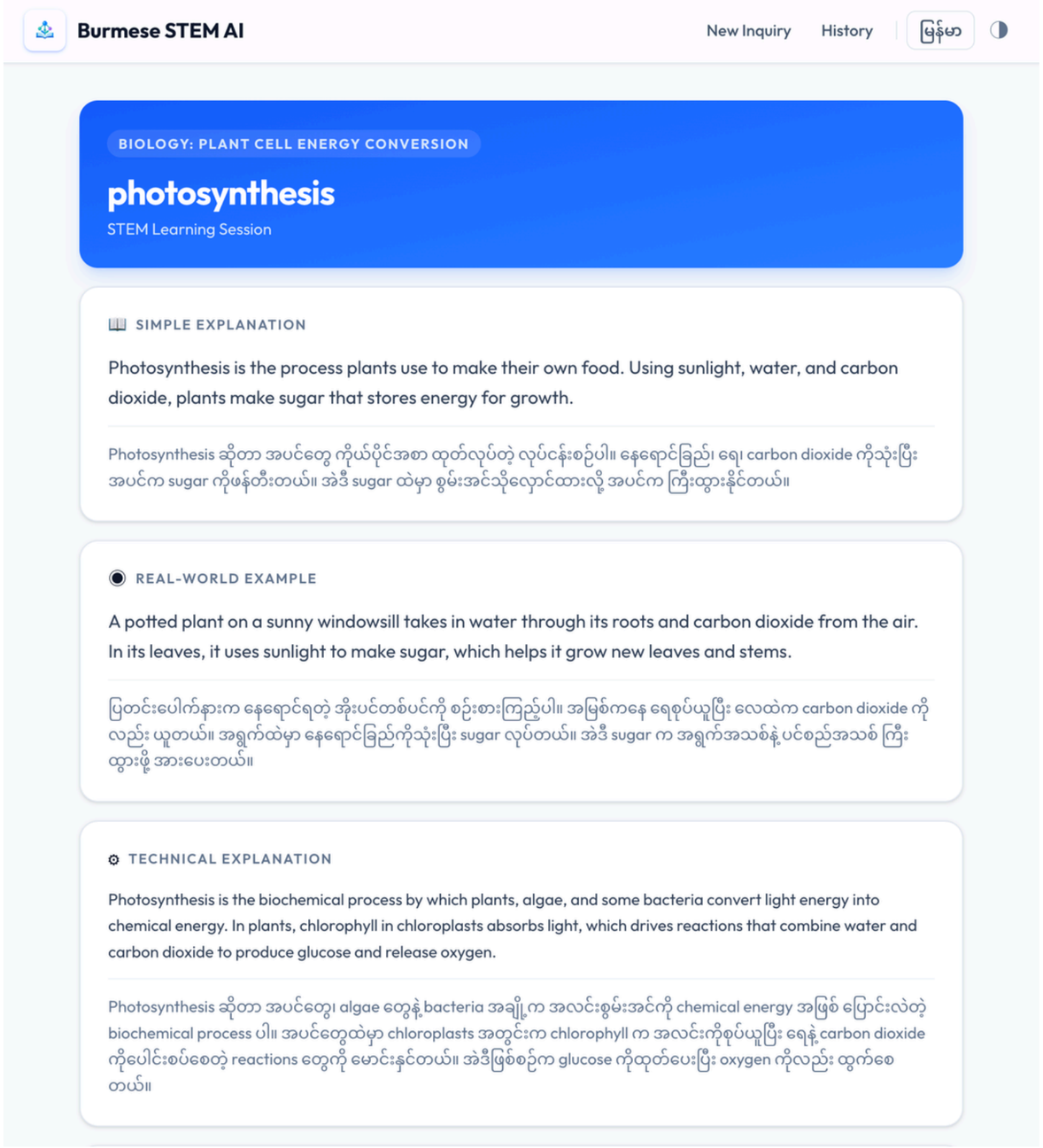

**System Artefact 3**

Burmese STEM AI prototype: Ask Question

Student asks a STEM question.

What is photosynthesis, and how do plants make food?

# DESIGNING: SYSTEM ARTEFACTS



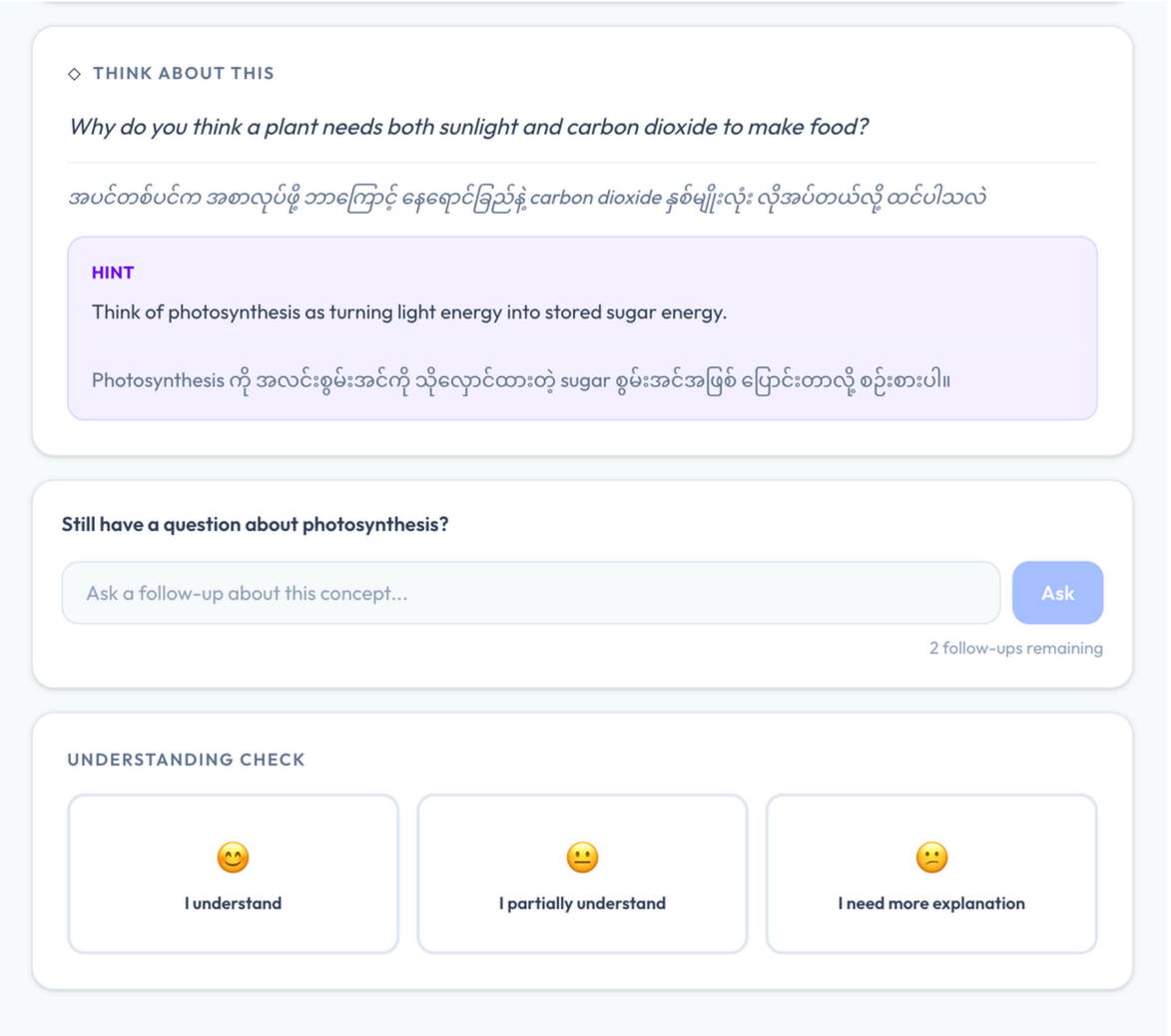

**System Artefact 3**

Burmese STEM AI prototype: Learning Session

Scaffold presents 3 bilingual explanations

- **Simple Explanation**
- **Real-World Example**
- **Technical Explanation**

# DESIGNING: SYSTEM ARTEFACTS



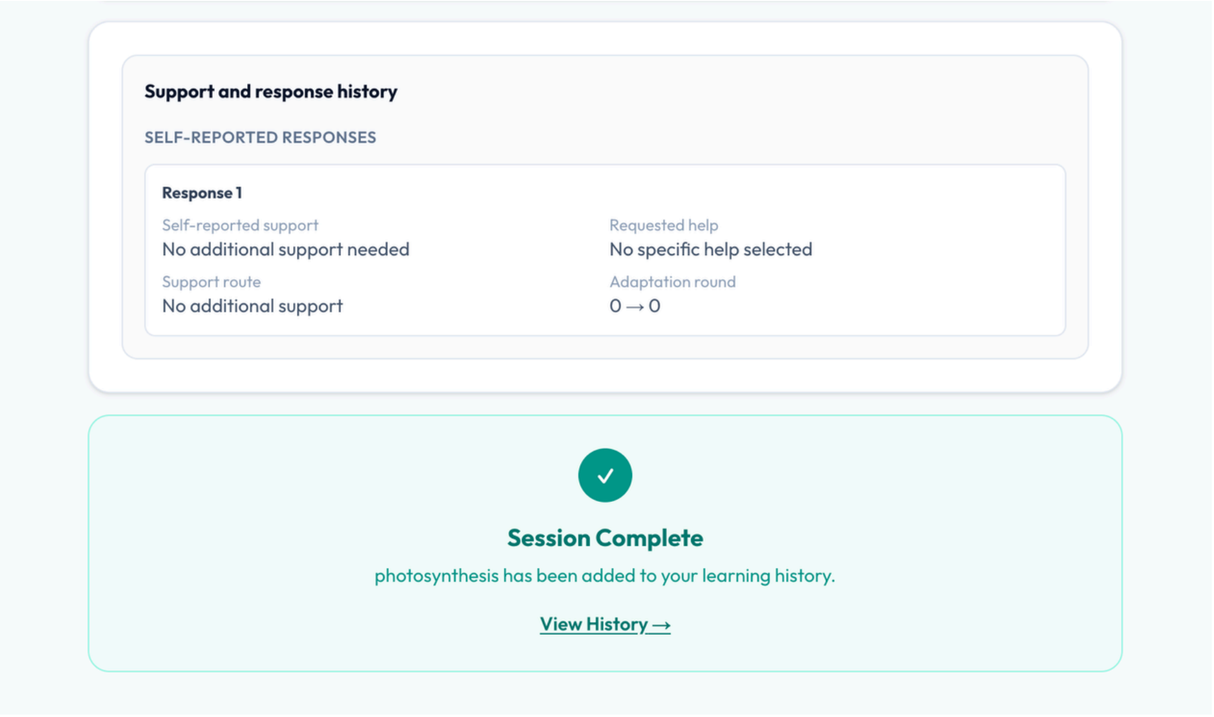

**System Artefact 3**

Burmese STEM AI prototype: Learning Session

Scaffold presents “**Think About This**” and “**Hint**” for reflection.

After reading everything, student responds “**Understanding Check**”

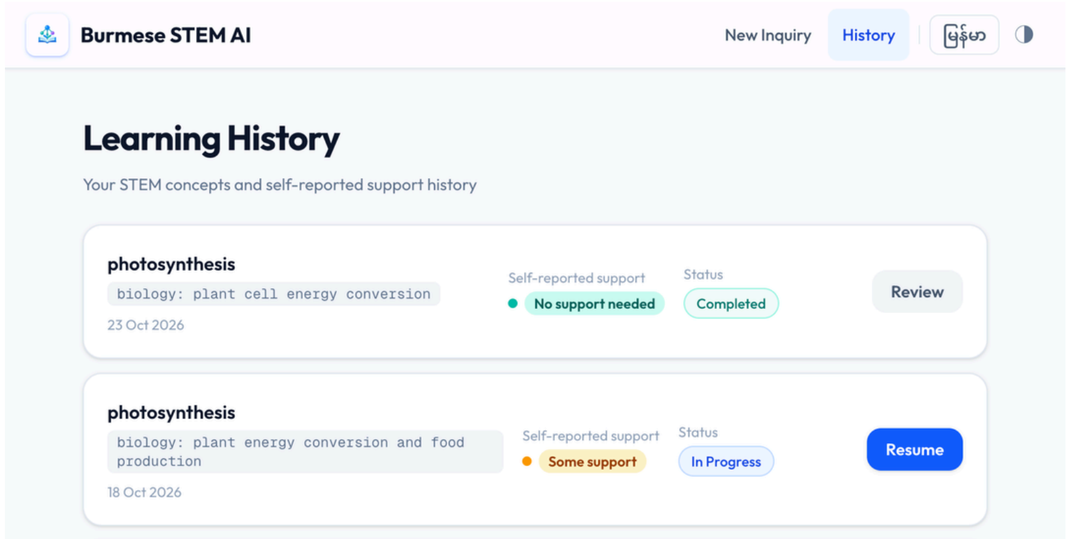
# DESIGNING: SYSTEM ARTEFACTS




‘**I understand**’ records **no further help** needed. Selecting **Finish** completes the session.

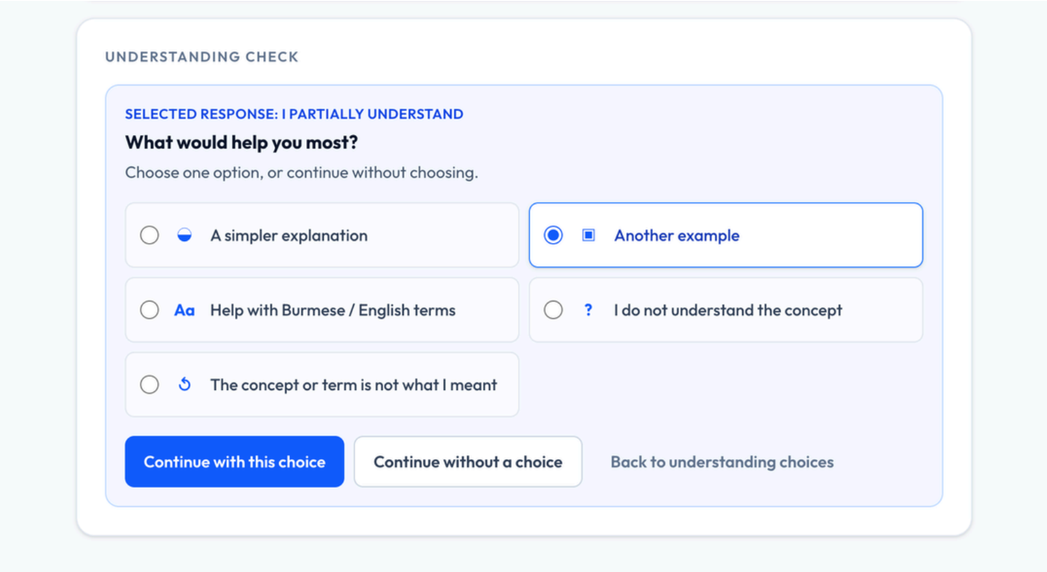
## System Artefact 3

Burmese STEM AI prototype: Learning Session




Learning history presents **completed learning session**.

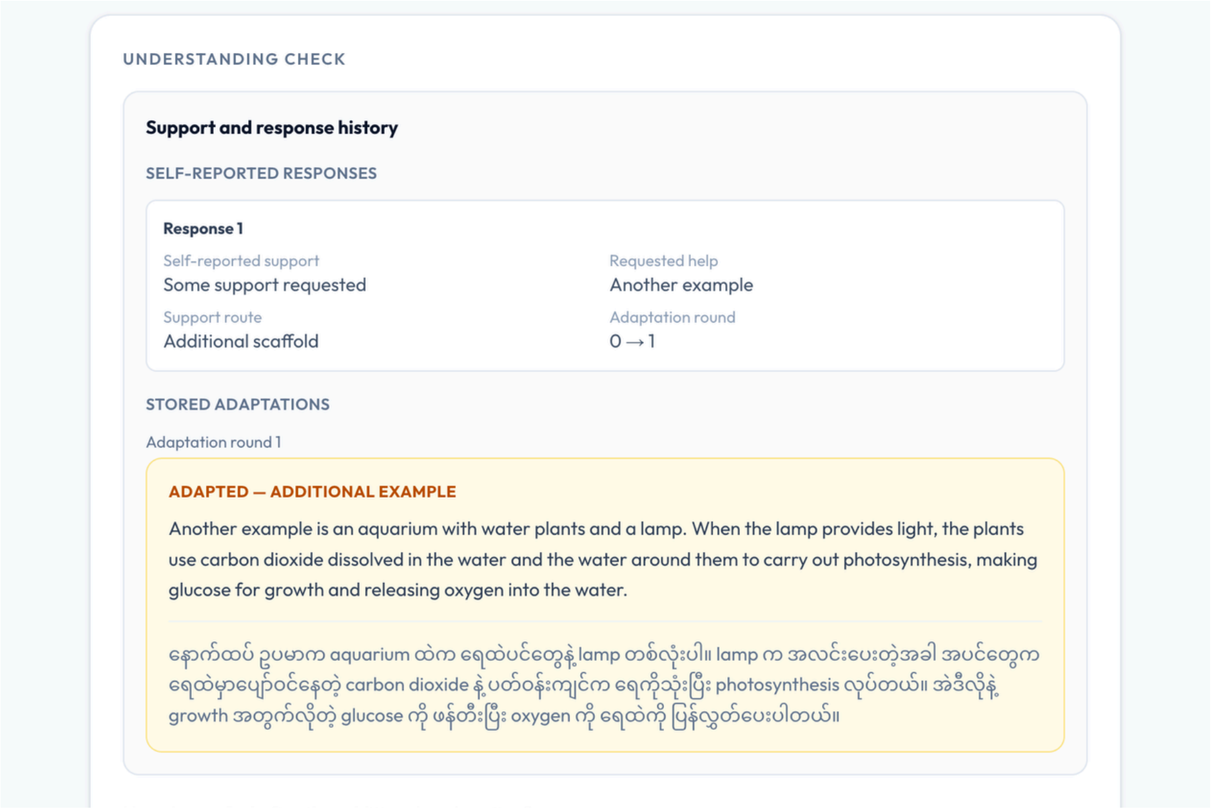
# DESIGNING: SYSTEM ARTEFACTS




If student responds with “**I partially understand**”, optional help choices will be asked.

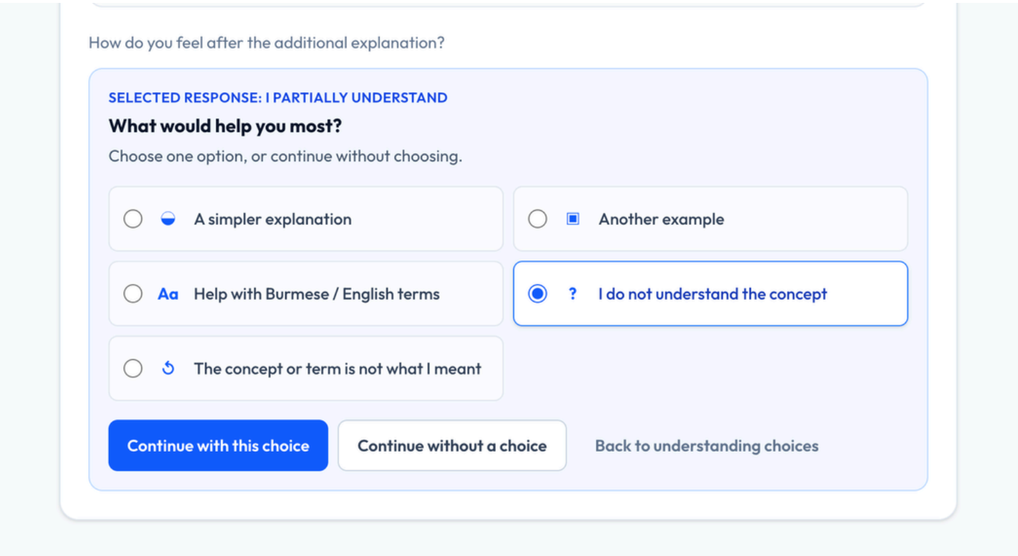
## System Artefact 3

Burmese STEM AI prototype: Learning Session




Scaffold provides “**another example**”.

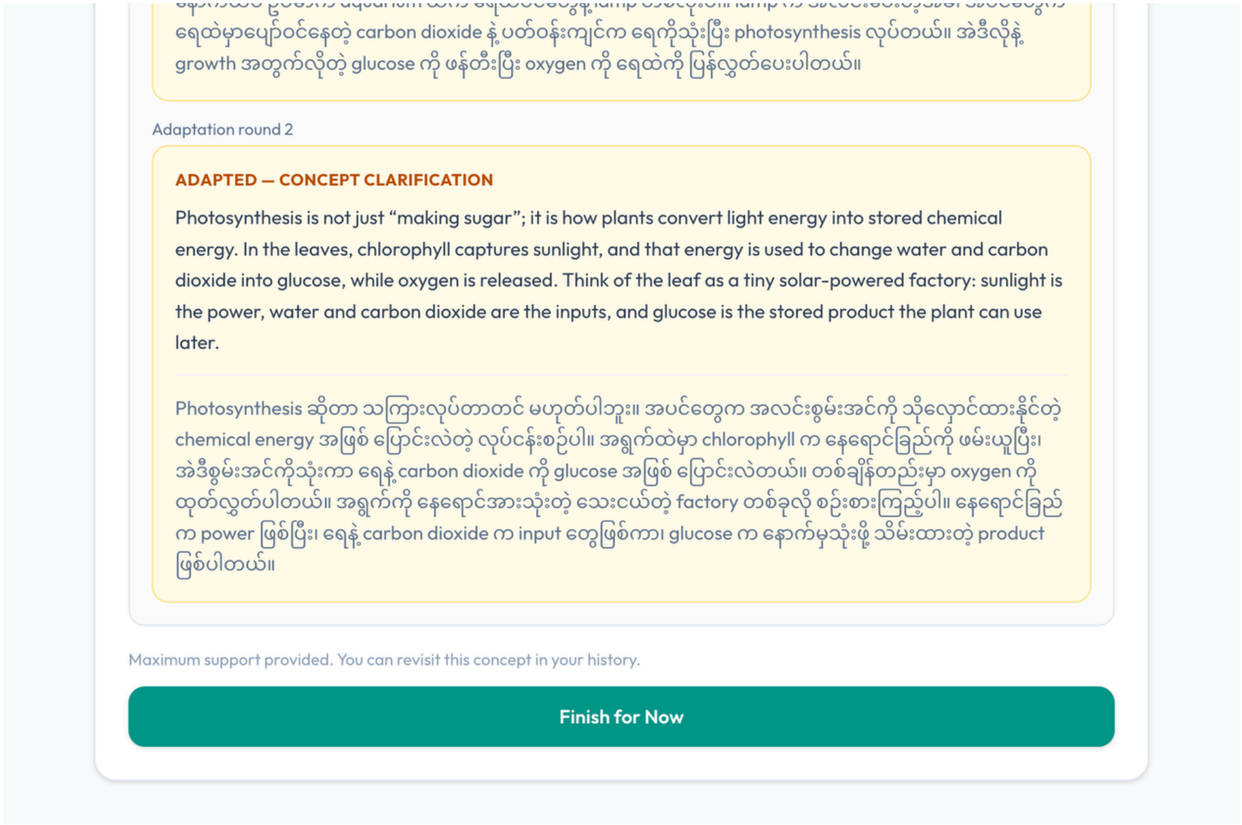
# DESIGNING: SYSTEM ARTEFACTS




Student asked again with “**I do not understand the concept**”.

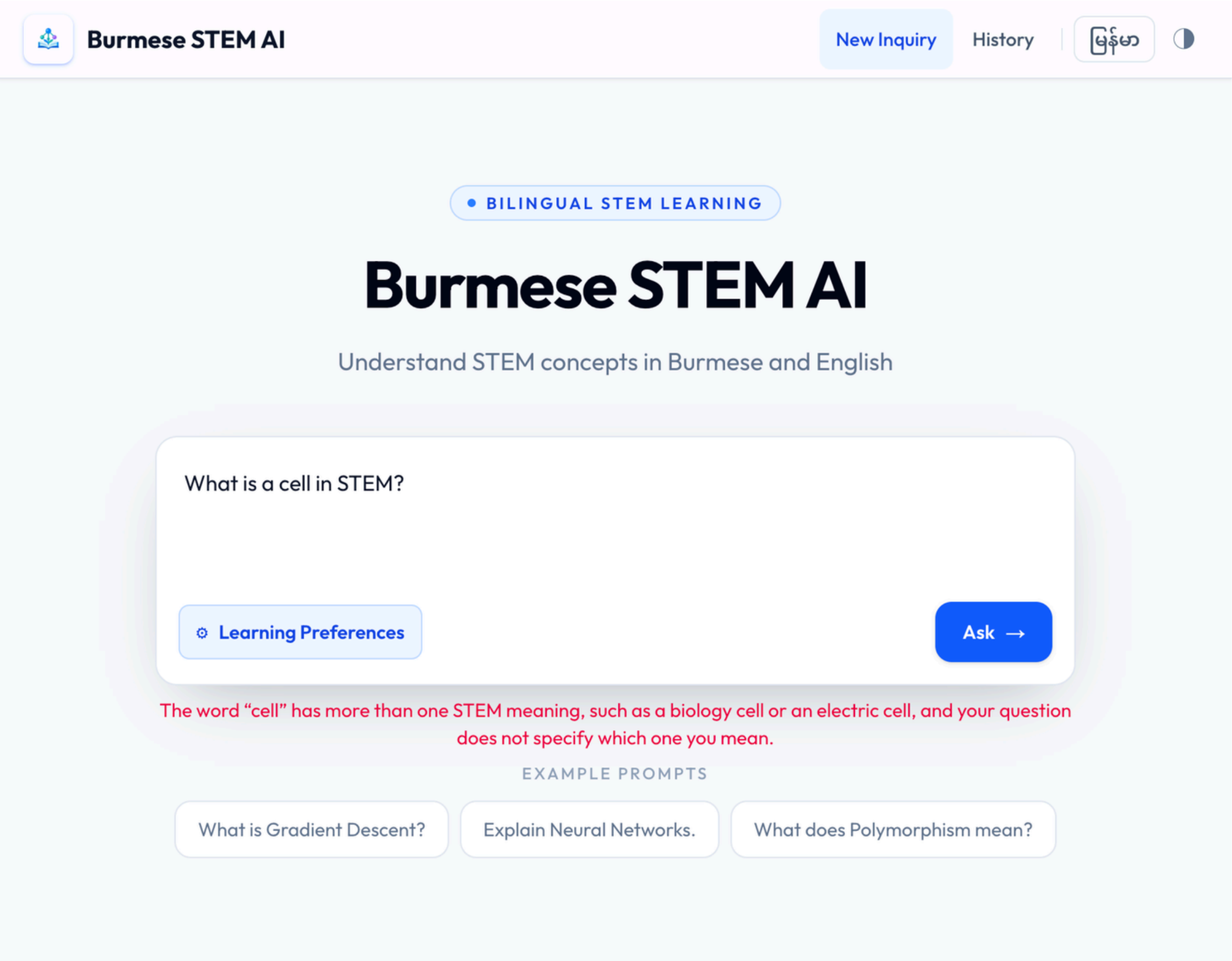
## System Artefact 3

Burmese STEM AI prototype: Learning Session




Scaffold provides “**concept clarification adaptation**”.
The limit of 2 saved support revisions has been reached.

# DESIGNING: SYSTEM ARTEFACTS




**System Artefact 3**

Burmese STEM AI prototype: Ask Question

The system detects the **ambiguous STEM term** and responds with error message.

# DESIGNING: PIRQOA TRACEABILITY

| P: Problem | I: Issue | R: Requirement | Q: Research question | O: Objective | A: Artefact link |
|---|---|---|---|---|---|
| • Language barriers to English STEM terms.<br>• Limited language resources and technology representation.<br>• Unfamiliar or misleading direct translations.<br>• Inconsistent STEM term identification. | Limited terminology support | Contextual bilingual help | RQ1: How can English STEM terms be supported? | Develop terminology support | **Framework**: Terms, context and language.<br><br>**Prototype**: Contextual Burmese/English terminology support |
| • Limited conceptual support from translation alone. | Limited conceptual support | Explanation and examples | RQ2: How can explanations go beyond translation? | Develop conceptual explanation | **Framework**: Explanation and scaffolding.<br><br>**Prototype**: Concept explanations, examples and hints. |
| • Limited personalised STEM terminology support.<br>• Insufficient guidance in unstructured LLM interaction. | Limited adaptive guidance | Structured, responsive help | RQ3: How can support be structured and adaptive? | Develop adaptive scaffolding | **Framework**: Response and adaptation.<br><br>**Prototype**: Help choices and bounded support revisions. |

OBSERVATION

THEORISING

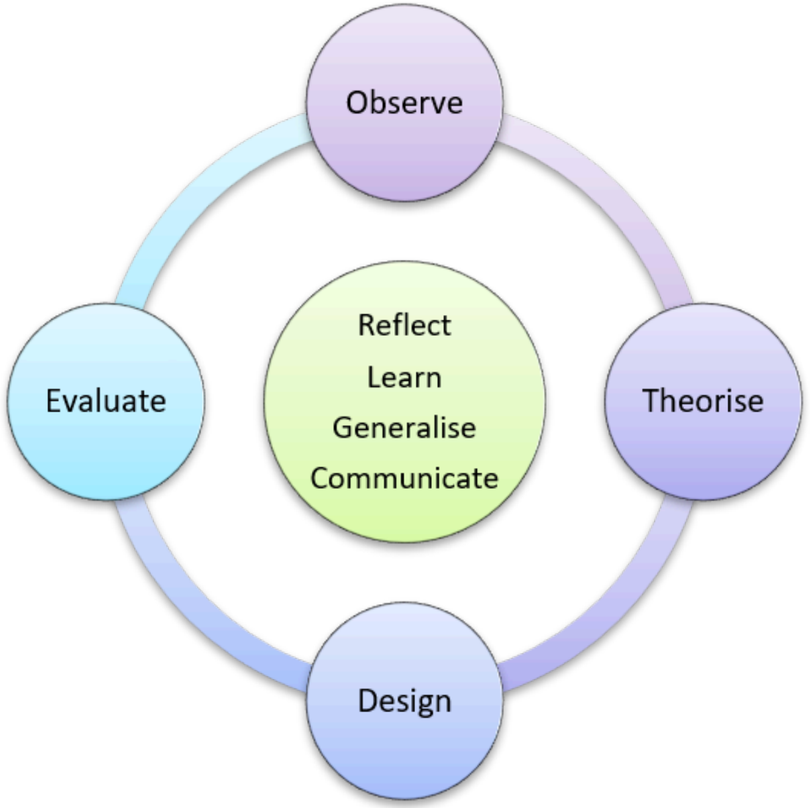
DESIGNING

**EVALUATION**

REFLECTION

GENERALISATION

CONTRIBUTION




Information System Research Process
(Gregor & Hevner, 2013)

# EVALUATION: CONCEPTUAL

## Four Conceptual Evaluation Methods

- Observational: **GenAI interview**
- SLR: Related to Academic **Literature**
- Descriptive: **Informed argument**
- Descriptive: Photosynthesis **scenario**

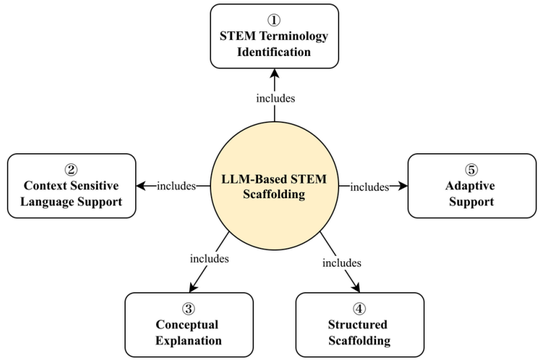
| | |
|---|---|
| 1. Observational | Case Study – Study artifact in depth in business environment |
| | Field Study – Monitor use of artifact in multiple projects |
| 2. Analytical | Static Analysis – Examine structure of artifact for static qualities (e.g., complexity) |
| | Architecture Analysis – Study fit of artifact into technical IS architecture |
| | Optimization – Demonstrate inherent optimal properties of artifact or provide optimality bounds on artifact behavior |
| | Dynamic Analysis – Study artifact in use for dynamic qualities (e.g., performance) |
| 3. Experimental | Controlled Experiment – Study artifact in controlled environment for qualities (e.g., usability) |
| | Simulation – Execute artifact with artificial data |
| 4. Testing | Functional (Black Box) Testing – Execute artifact interfaces to discover failures and identify defects |
| | Structural (White Box) Testing – Perform coverage testing of some metric (e.g., execution paths) in the artifact implementation |
| 5. Descriptive | Informed Argument – Use information from the knowledge base (e.g., relevant research) to build a convincing argument for the artifact's utility |
| | Scenarios – Construct detailed scenarios around the artifact to demonstrate its utility |

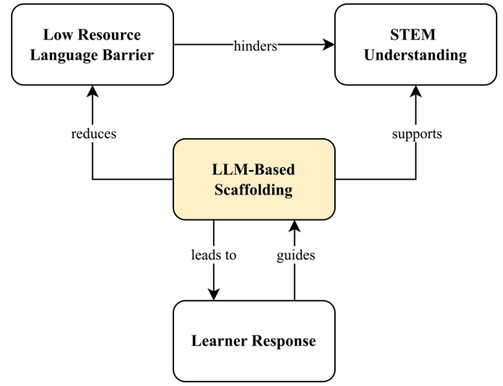
Evaluation Methods
(Hevner et al., 2004)

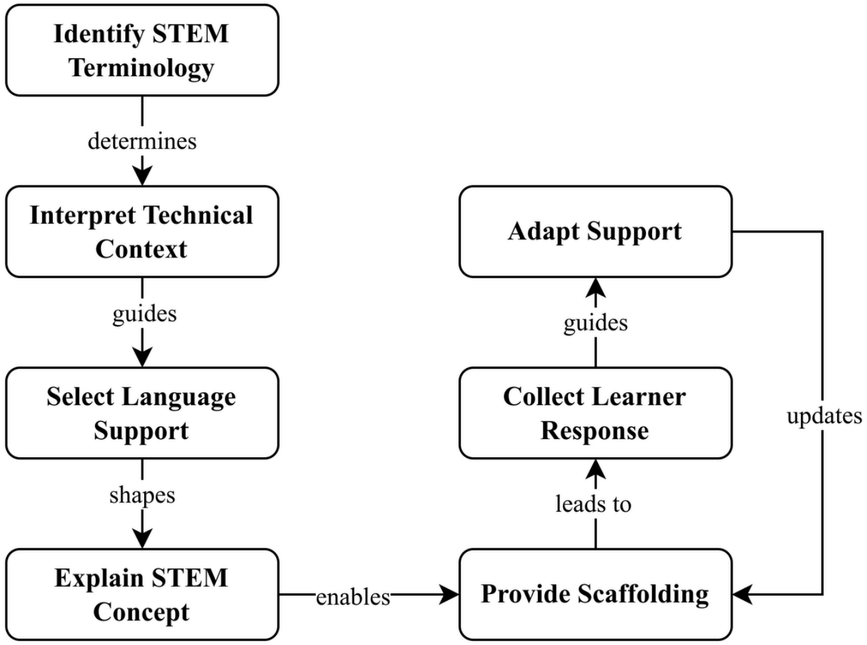
# EVALUATION: CONCEPTUAL

## 1st Evaluation Results

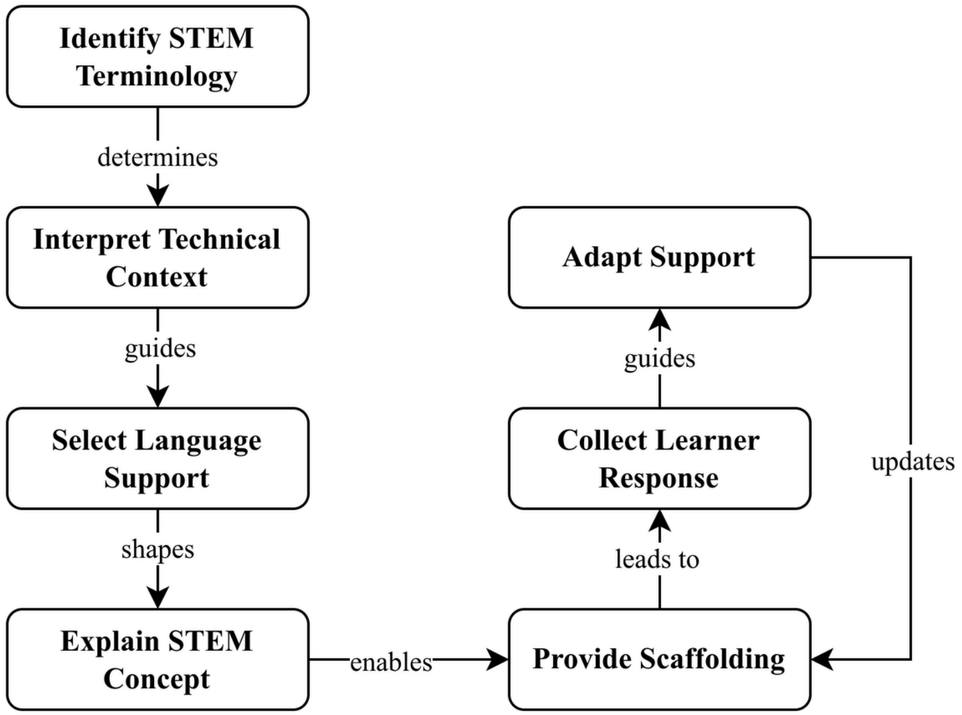
- All main support steps are included
- Support rules need clarification
  - **Stage roles**
  - **Language choice**
  - **Feedback decisions**



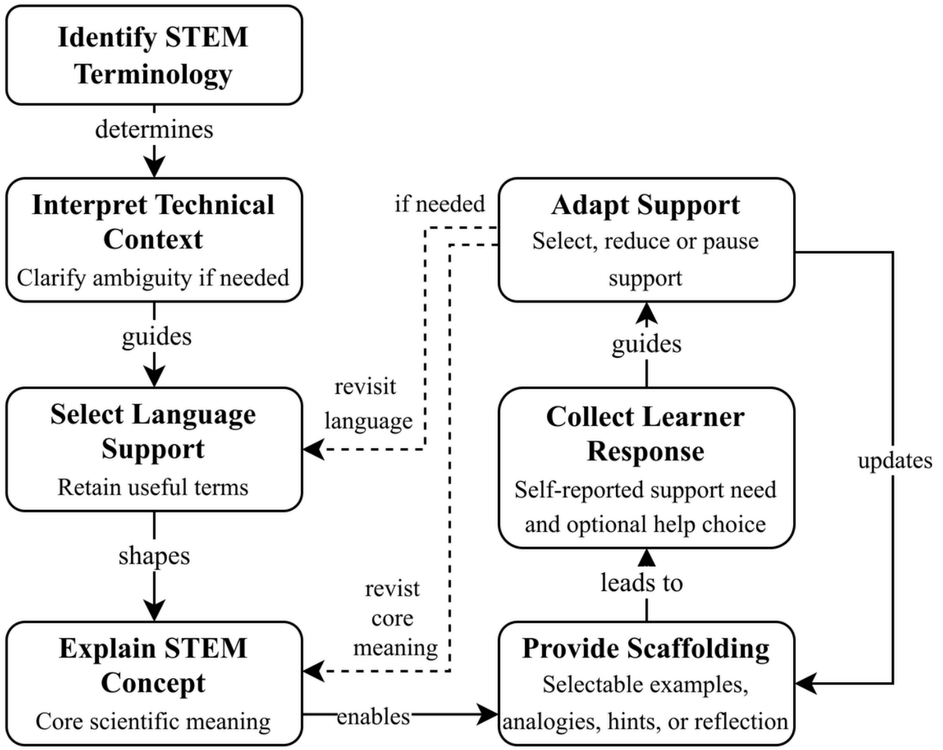



# EVALUATION: CONCEPTUAL




Conceptual Artefact 3 (Original)



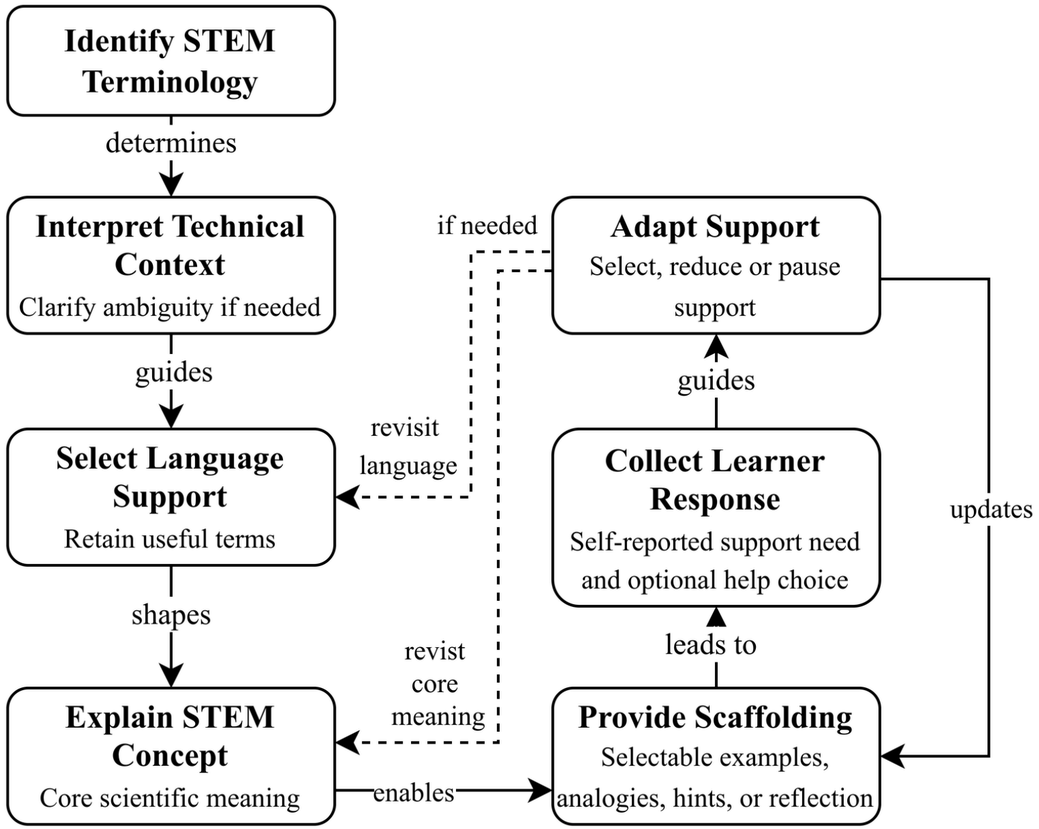

Conceptual Artefact 3 (After Refinement)

- Core meaning and extra help distinguished
- Self-reported support need made explicit
- Routes to Language and core-meaning feedback added

# EVALUATION: CONCEPTUAL

## 2nd Evaluation Iteration

- Required support is covered
- Some decisions remain unclear
- Learning benefits are unverified



# EVALUATION: *SYSTEM*

## System Evaluation Methods

- Software checks
  - **Static** and **dynamic** analysis
  - **Black-box** and **white-box** testing
  - **Bounds** and concurrency analysis
- Content and usability checks
  - **Simulation** and content review
  - Structured **usability** inspection
- Design checks
  - **Literature comparison**
  - **Informed argument**
  - Executed photosynthesis **scenario**

| | |
|---|---|
| 1. Observational | Case Study – Study artifact in depth in business environment |
| | Field Study – Monitor use of artifact in multiple projects |
| 2. Analytical | Static Analysis – Examine structure of artifact for static qualities (e.g., complexity) |
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| 5. Descriptive | Informed Argument – Use information from the knowledge base (e.g., relevant research) to build a convincing argument for the artifact's utility |
| | Scenarios – Construct detailed scenarios around the artifact to demonstrate its utility |

Evaluation Methods
(Hevner et al., 2004)

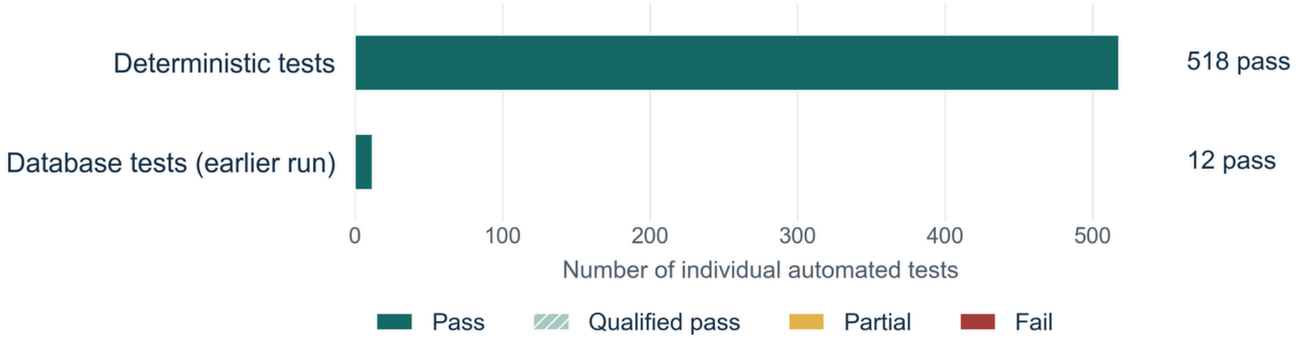
# EVALUATION: SYSTEM

## Test Results




- 2 dynamic partials: LLM requests were not directly counted.
- 2 black-box partials: Extra help worked, but its usefulness was not checked.

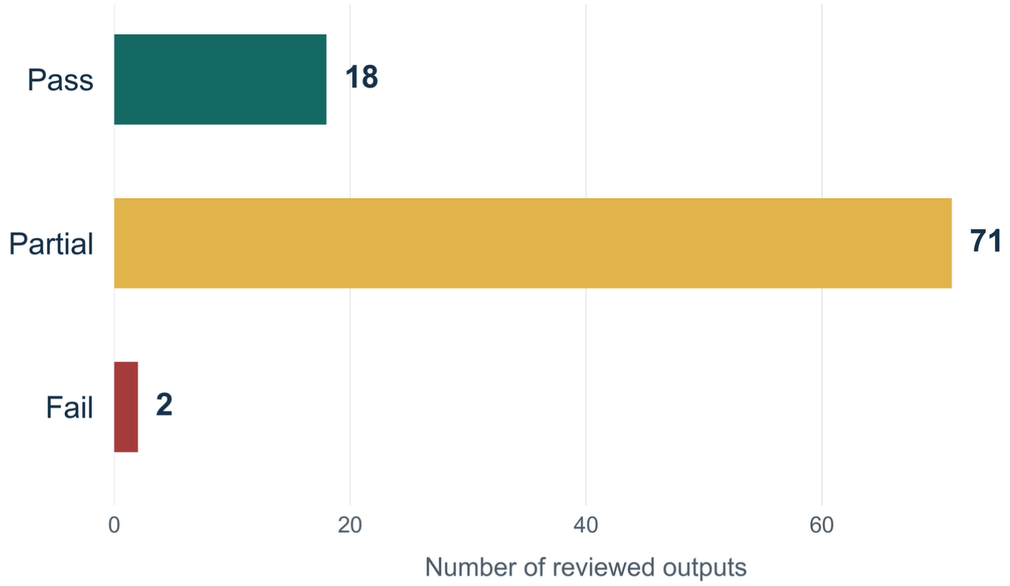
## White-box Testing




- 1 black-box fail: Test expected error but system created anonymous identity.

# EVALUATION: SYSTEM

## Simulation Test Results

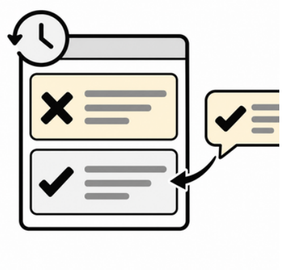



Author review; no independent second assessor

- **Main finding**: Most outputs needed improvement.
- **Partial**: Missing detail, unclear Burmese or repeated explanations.
- **Fail**: Errors changed the scientific meaning.

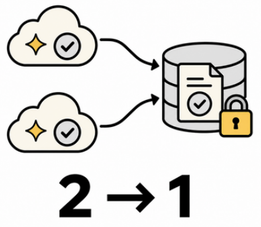
- **Example**
  - **gravity (Fail)**: Mass (ဒြပ်ထု) was translated as weight (အလေးချိန်).
  - **photosynthesis (Partial)**: “အစားအစာ 만들ဖို့” mixed Korean into Burmese. Suggested wording: “အစားအစာထုတ်လုပ်ဖို့”.

# EVALUATION: SYSTEM



## Simulation Test reveals

Even after further help, old mistakes remain in saved history




## Bound Test reveals

2 Generation Calls → 1 Saved Answer

- A limit of 2 saved revisions does not limit LLM calls.



## Usability Inspection reveals

Some interface messages and controls need improvement

- Usability inspection: 6 criteria passed; 3 were partly met.
- Example: error messages stayed in English when the interface was set to Burmese.

OBSERVATION

THEORISING

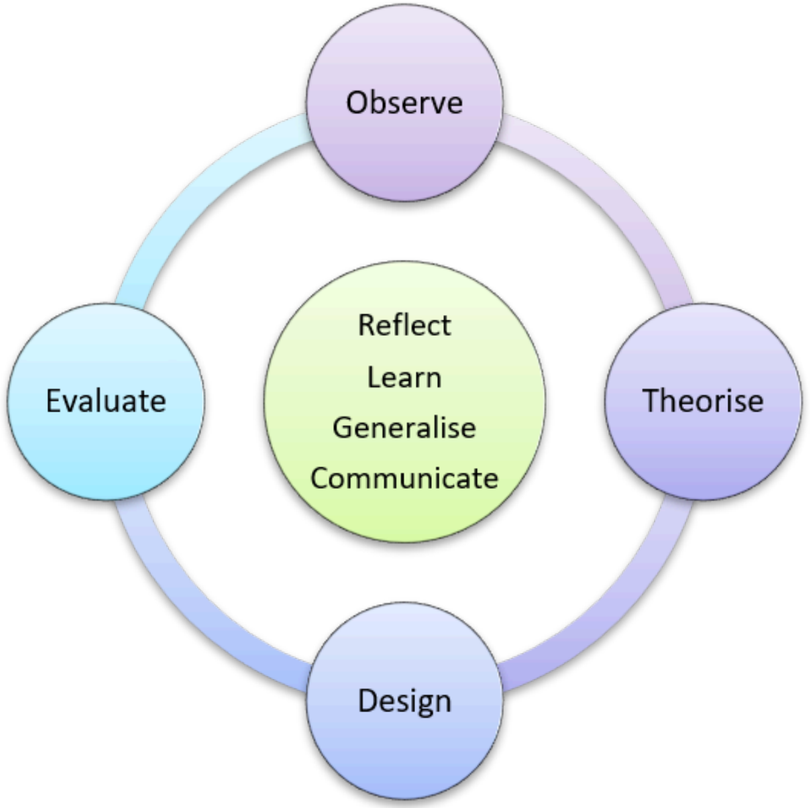
DESIGNING

EVALUATION

**REFLECTION**

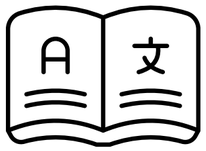
GENERALISATION

CONTRIBUTION




Information System Research Process
(Gregor & Hevner, 2013)

# REFLECTION



## Valid format ≠ Correct meaning

- Passing software tests does not prove correct science or Burmese meaning.
- Educational claims need different evidence.



## More help ≠ Better help

- Giving more examples may not address their actual difficulty.
- Test with learners whether the selected help actually addresses their difficulty.



## Saved limits ≠ Spending limits

- Saved limits do not guarantee LLM spending limits.
- Control saved answers and generation requests separately.

OBSERVATION

THEORISING

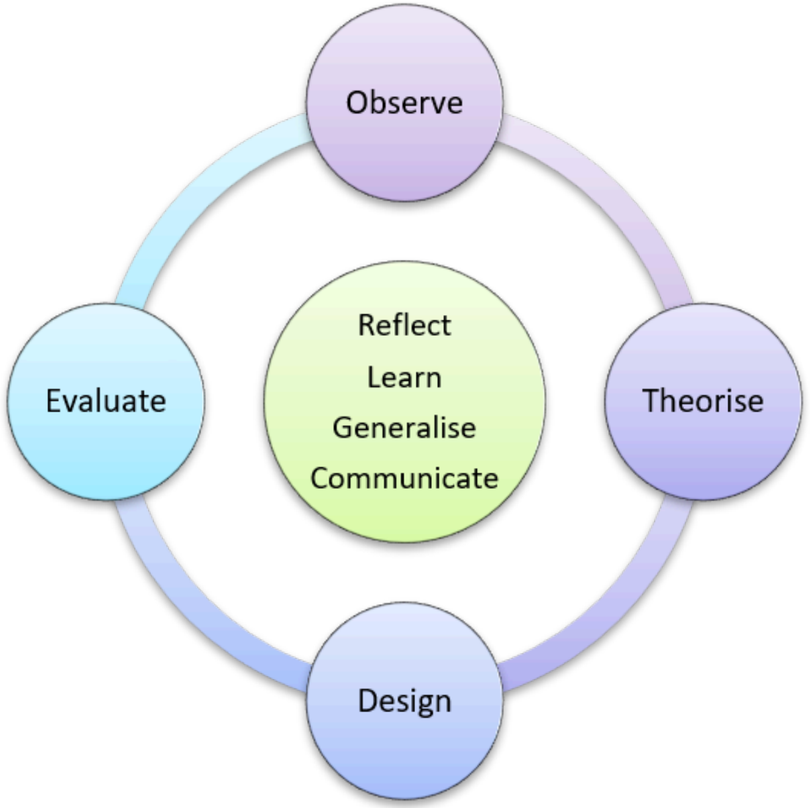
DESIGNING

EVALUATION

REFLECTION

**GENERALISATION**

CONTRIBUTION




Information System Research Process
(Gregor & Hevner, 2013)

# GENERALISATION



## Design Principle 1

- Clarify technical meaning before choosing language support.



## Design Principle 2

- Use learner requests to guide help, not prove understanding.



## Design Principle 3

- Check scientific accuracy, bilingual consistency and response structure separately

# GENERALISATION



## Design Principle 4

- Record what changed, what remains unclear, what help was requested, and whether the session is complete.



## Design Principle 5

- Distinguish limits on saved revisions, model calls, time and cost

OBSERVATION

THEORISING

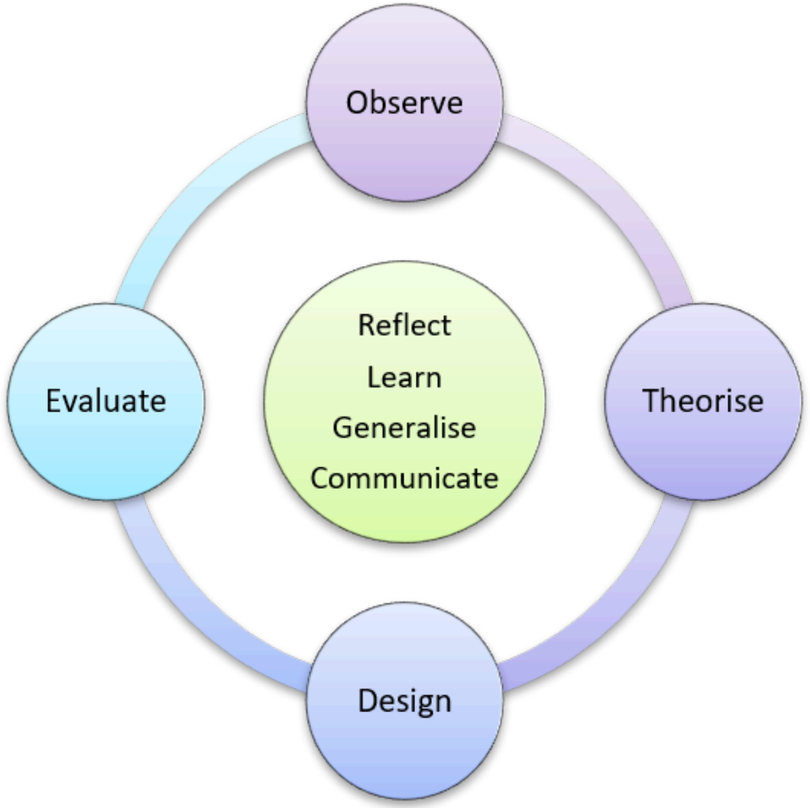
DESIGNING

EVALUATION

REFLECTION

GENERALISATION

**CONTRIBUTION**




Information System Research Process
(Gregor & Hevner, 2013)

# CONTRIBUTION

**Intended contribution**: Improvement

(Gregor & Hevner, 2013)




- Builds on TTF and Scaffolding Theory. Better learning outcomes remain untested.
- Combines STEM terminology support, concept explanations and learner-requested help.
- Guide support decisions, content checks, session records and interaction limits. These need further testing.

# CONTRIBUTION — LIMITATIONS AND FUTURE WORK



## No independent content assessment

- Obtain bilingual Burmese-speaking STEM specialist review



## One setting

- Test other languages and support limits



## No learner study

- Compare understanding and retention

Independently verify explanations → Compare learner outcomes with a baseline → Evaluate support/cost limits → Test transfer to another language and subject

# Conclusion

# CONCLUSION



## RQ1: Language Barrier

How can specialized English STEM terminology be supported for Burmese-speaking learners?

- [Research] System provided **contextual Burmese support** for English STEM terms.
- [Limit] Consistent translation accuracy and reduced language difficulty remain unverified.



## RQ2: Conceptual Difficulties

How can LLM-based support help learners understand STEM concepts beyond translation?

- [Research] System generated **explanations**, **examples**, **hints** and **reflection** questions.
- [Limit] Improved understanding and retention were not measured.



## RQ3: Adaptive Guidance

How can LLM-based scaffolding provide structured and adaptive support for Burmese-speaking STEM learners?

- [Research] Help choices **guided responses**, and tested **support limits** worked
- [Limit] Whether the selected help met learners' needs remains unverified.

# Thank You

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- Grateful to Rizky Andrian for his guidance on the research process.

## Scan for Live Demo



https://burmesestemai.serveirc.com/



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